

EE115B *Digital Circuits*

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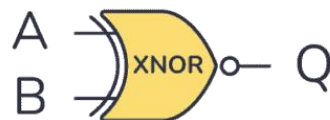
Combinational Logic

Comparators

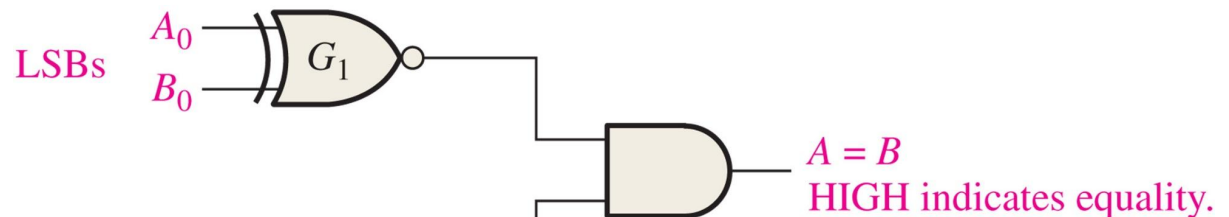
Mux

Comparators (Equality)

The function of a comparator is to compare the magnitudes of two binary numbers to determine the relationship between them. In the simplest form, a comparator can test for equality using XNOR gates.



A	B	Q
0	0	1
0	1	0
1	0	0
1	1	1



General format: Binary number $A \rightarrow A_1A_0$
Binary number $B \rightarrow B_1B_0$



Can you design a circuit to test two four-bit numbers for equality?



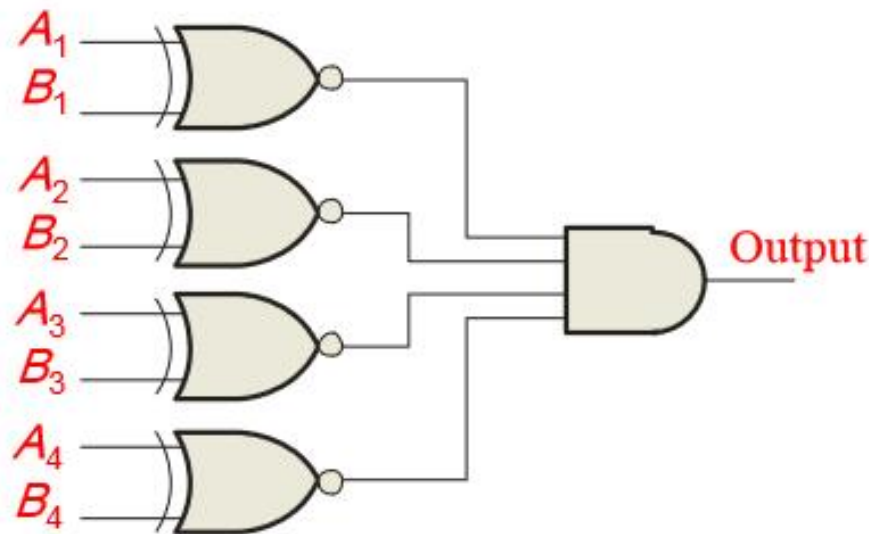
Exercise

The function of a comparator is to compare the magnitudes of two binary numbers to determine the relationship between them. In the simplest form, a comparator can test for equality using XNOR gates.

Example Solution

How could you test two 4-bit numbers for equality?

AND the outputs of four XNOR gates.



1-bit Numerical Comparators

The function of a comparator is to compare the magnitudes of two binary numbers to determine the relationship between them. In the simplest form, a comparator can test for equality using XNOR gates.

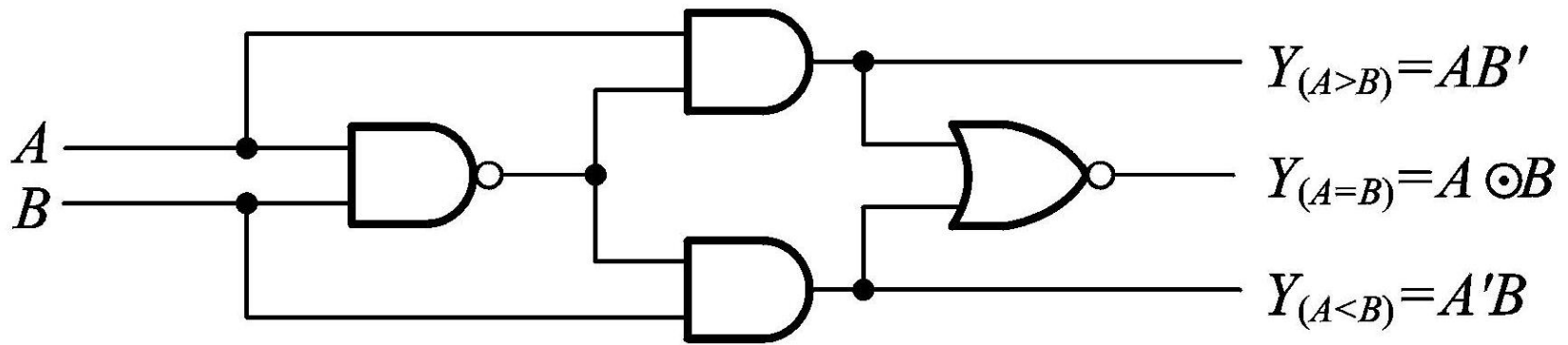


Diagram of 1 bit comparator

Multiple-bit Numerical Comparator

Multiple bits: Compare from high bit to low bit

C and D: numbers with n+1 bits (general expressions)

1. **For $Y(C > D)$:**

$$Y(C > D) = C_n \overline{D_n} + (C_n \odot D_n) C_{n-1} \overline{D_{n-1}} + \cdots + (C_n \odot D_n) (C_{n-1} \odot D_{n-1}) \cdots (C_1 \odot D_1) C_0 \overline{D_0}$$

2. **For $Y(C < D)$:**

$$Y(C < D) = \overline{C_n} D_n + (C_n \odot D_n) \overline{C_{n-1}} D_{n-1} + \cdots + (C_n \odot D_n) (C_{n-1} \odot D_{n-1}) \cdots (C_1 \odot D_1) \overline{C_0} D_0$$

3. **For $Y(C = D)$:**

$$Y(C = D) = (C_n \odot D_n) (C_{n-1} \odot D_{n-1}) \cdots (C_1 \odot D_1) (C_0 \odot D_0)$$

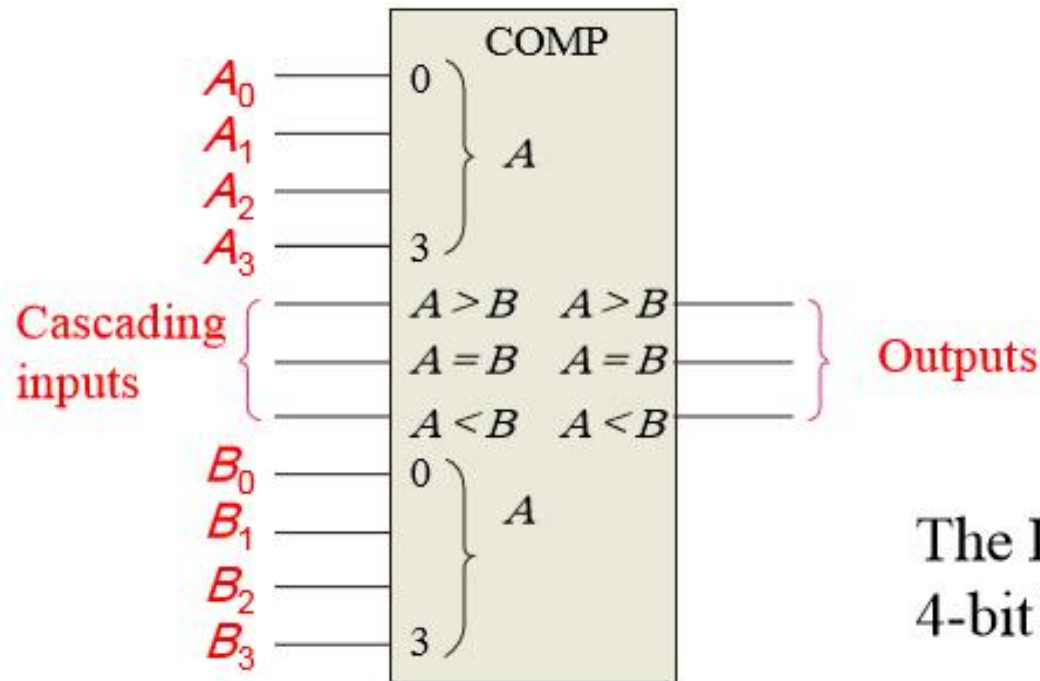
Does this circuit suffer from long propagation delay?

How many XNOR gate it needs?



Numerical Comparator (Cascaded)

Assume you have a four-bit comparator. How can you build a 8 bit comparator?



The IC shown is the 4-bit 74LS85.



Numerical Comparator (Cascaded)

1. **A and B: 4 highest-order bits of numbers with ≥ 4 bits**
2. **Results of lower-order bits (carry): $I_{(A>B)}$, $I_{(A<B)}$, and $I_{(A=B)}$**
3. **For numbers with ≤ 4 bits (first stage of cascaded structure):**
 $I_{(A>B)}=I_{(A<B)}=0$, $I_{(A=B)}=1$

1. **For $Y(A > B)$:**

$$\begin{aligned} Y(A > B) = & A_3 B'_3 + (A_3 \odot B_3) A_2 B'_2 + (A_3 \odot B_3) (A_2 \odot B_2) A_1 B'_1 \\ & + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 B'_0 \\ & + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0) I_{(A>B)} \end{aligned}$$

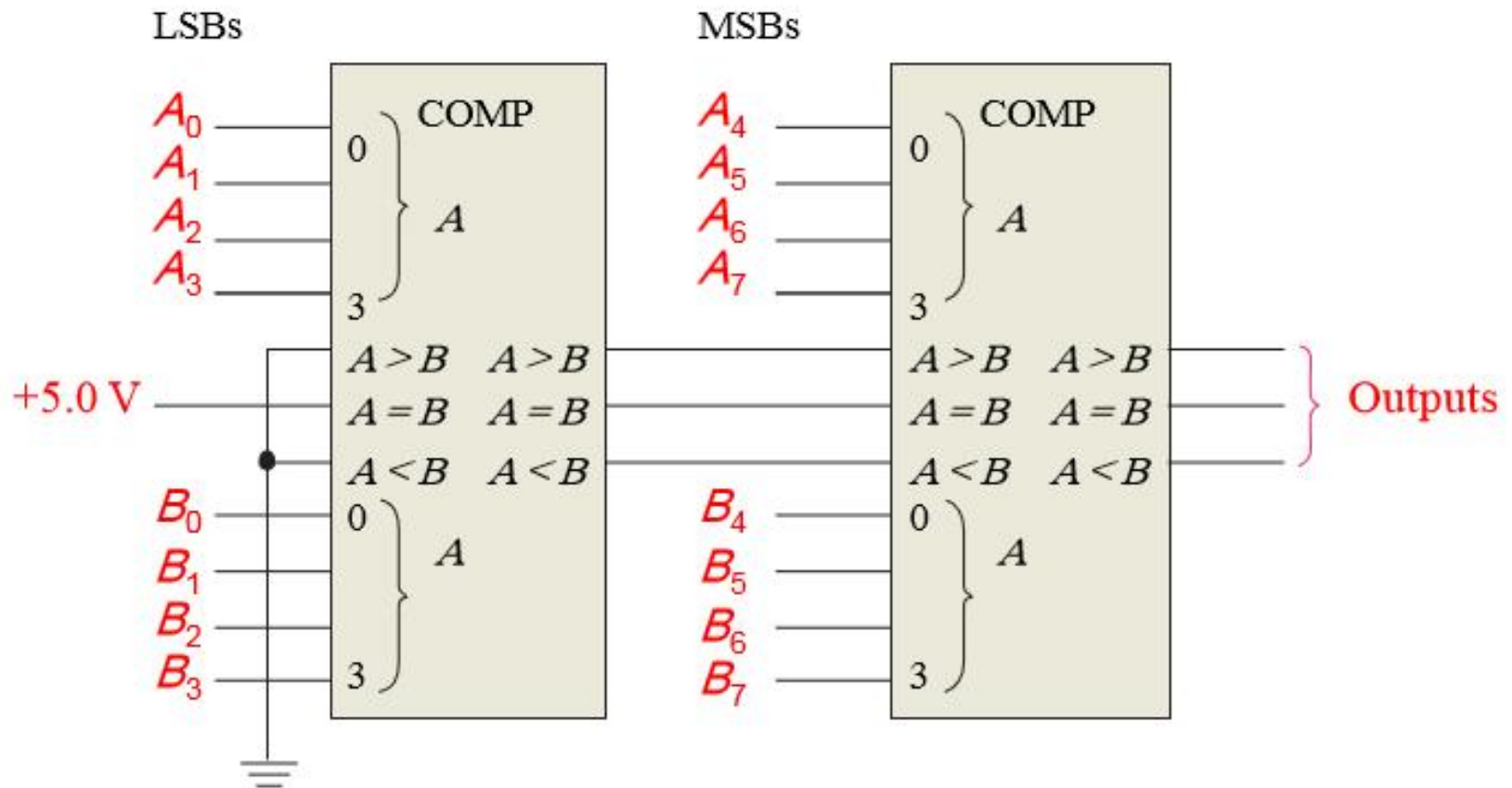
2. **For $Y(A < B)$:**

$$\begin{aligned} Y(A < B) = & A'_3 B_3 + (A_3 \odot B_3) A'_2 B_2 + (A_3 \odot B_3) (A_2 \odot B_2) A'_1 B_1 \\ & + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A'_0 B_0 \\ & + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0) I_{(A<B)} \end{aligned}$$

Can you write $Y(A=B)$?

Numerical Comparator (Cascaded)

IC comparators can be expanded using the cascading inputs as shown. The lowest order comparator has a HIGH on the $A = B$ input.



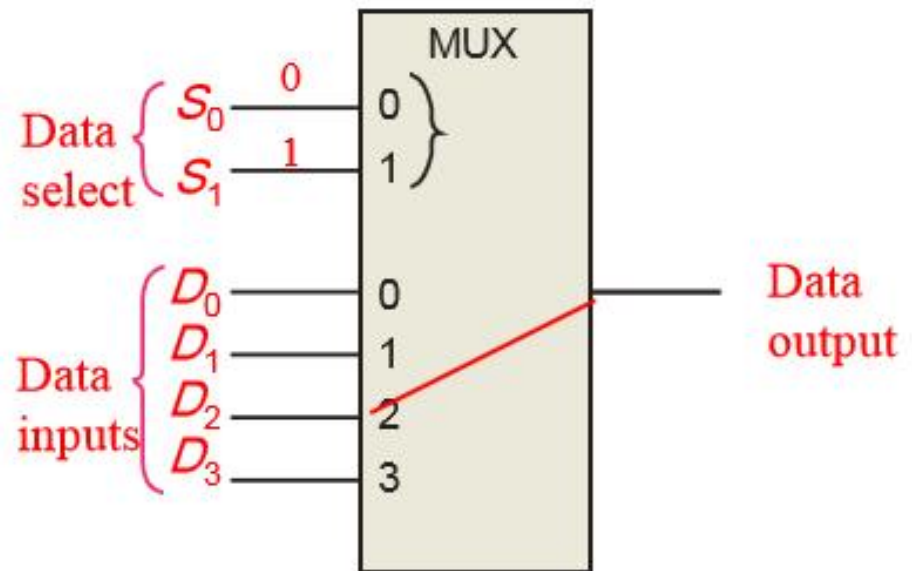
Multiplexers (Data Selectors)

A multiplexer (MUX) selects one data line from two or more input lines and routes data from the selected line to the output. The particular data line that is selected is determined by the select inputs.

Two select lines are shown here to choose any of the four data inputs.

Question

Which data line is selected if $S_1 S_0 = 10$? D_2



4-to-1 (1-of-4) MUX

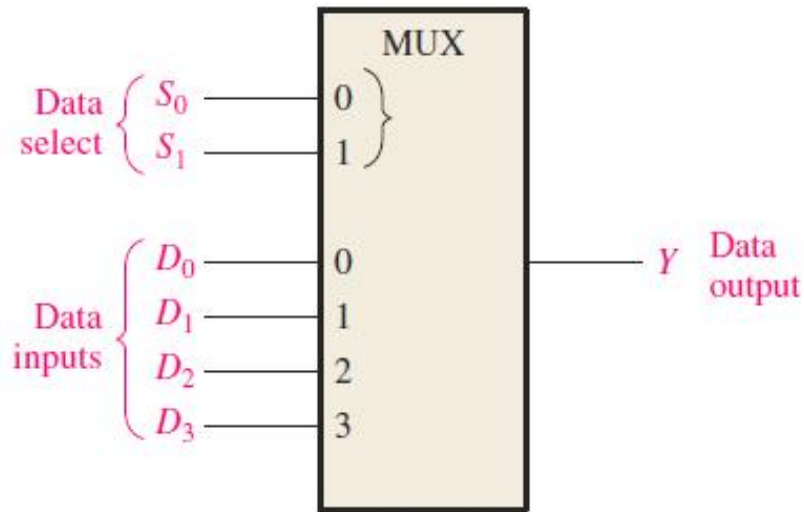


TABLE 6-8

Data selection for a 1-of-4-multiplexer.

Data-Select Inputs		Input Selected
S_1	S_0	
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

The data output is equal to D_0 only if $S_1 = 0$ and $S_0 = 0$: $Y = D_0 \bar{S}_1 \bar{S}_0$.

The data output is equal to D_1 only if $S_1 = 0$ and $S_0 = 1$: $Y = D_1 \bar{S}_1 S_0$.

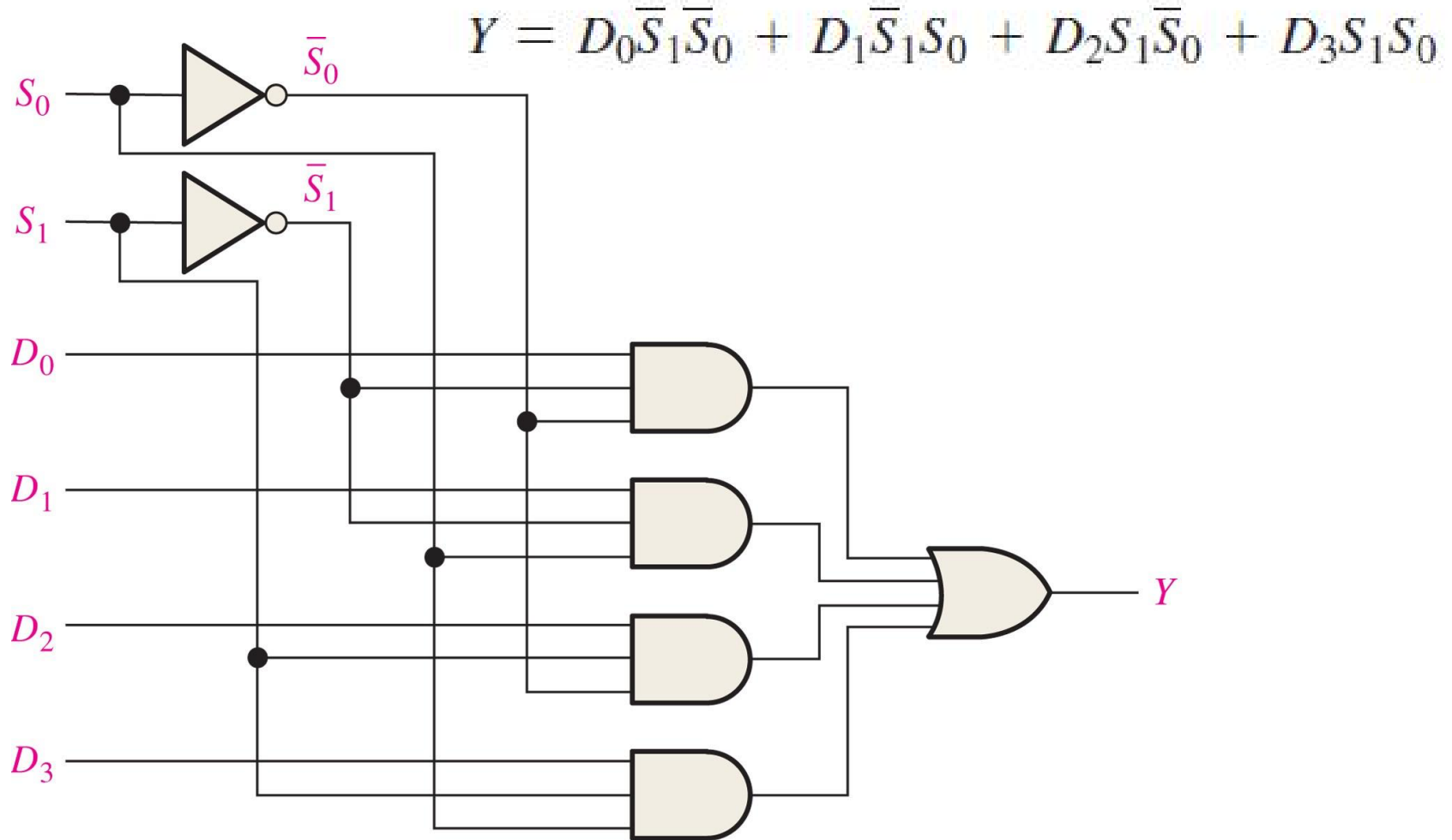
The data output is equal to D_2 only if $S_1 = 1$ and $S_0 = 0$: $Y = D_2 S_1 \bar{S}_0$.

The data output is equal to D_3 only if $S_1 = 1$ and $S_0 = 1$: $Y = D_3 S_1 S_0$.

When these terms are ORed, the total expression for the data output is

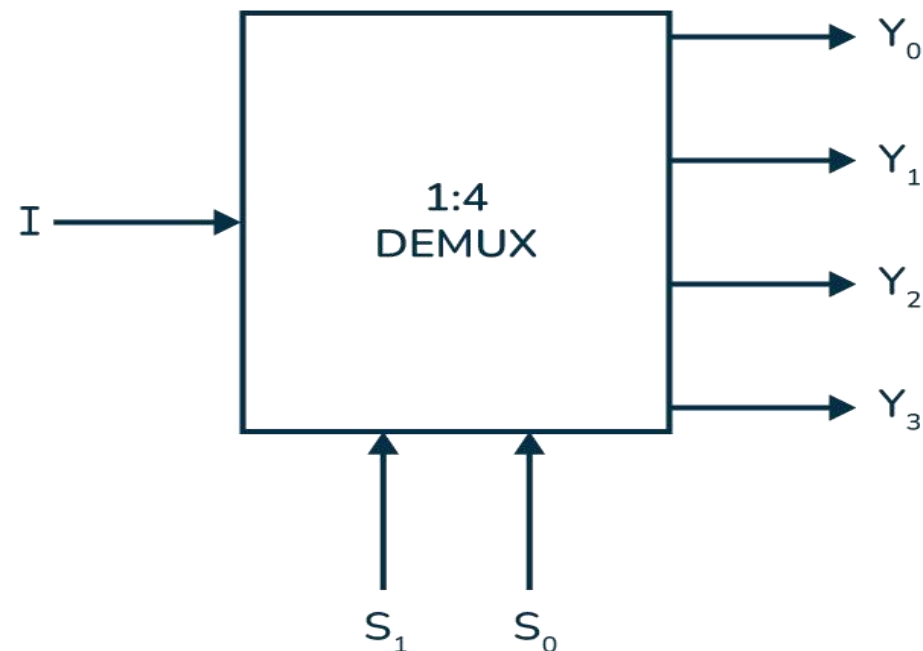
$$Y = D_0 \bar{S}_1 \bar{S}_0 + D_1 \bar{S}_1 S_0 + D_2 S_1 \bar{S}_0 + D_3 S_1 S_0$$

4-to-1 (1-of-4) MUX



Demultiplexers (Data Distributors)

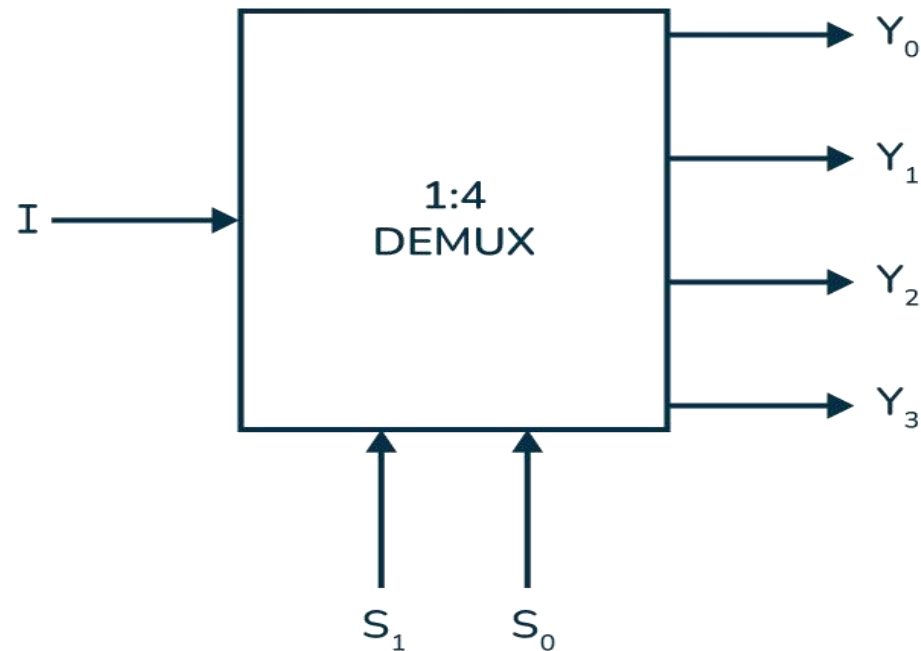
A demultiplexer (DEMUX) performs the opposite function from a MUX. It switches data from one input line to two or more data lines depending on the select inputs.



Demultiplexers (Data Distributors)

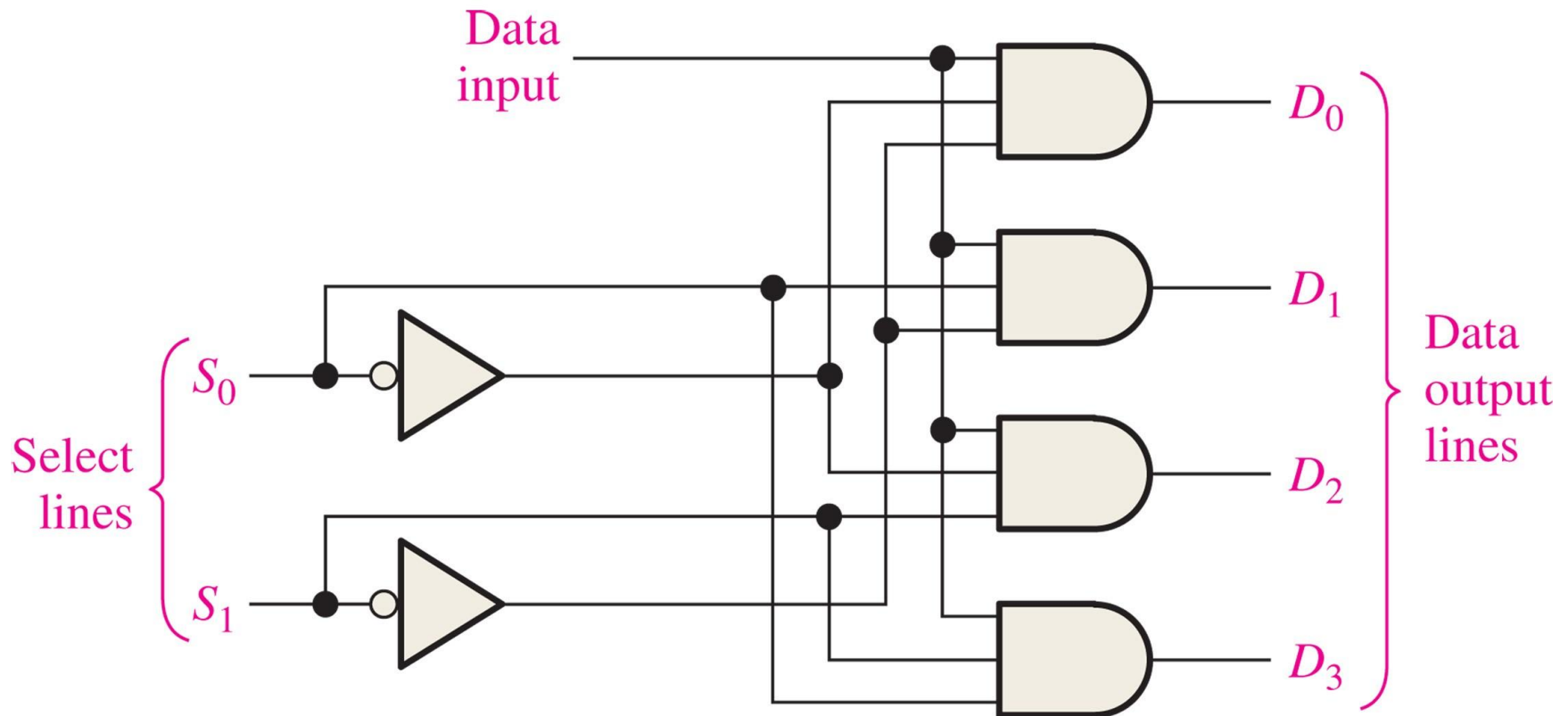
Exercise: Design the circuit of 4 bit demultiplexer

Hint: Think about let whom to pass using a AND gate



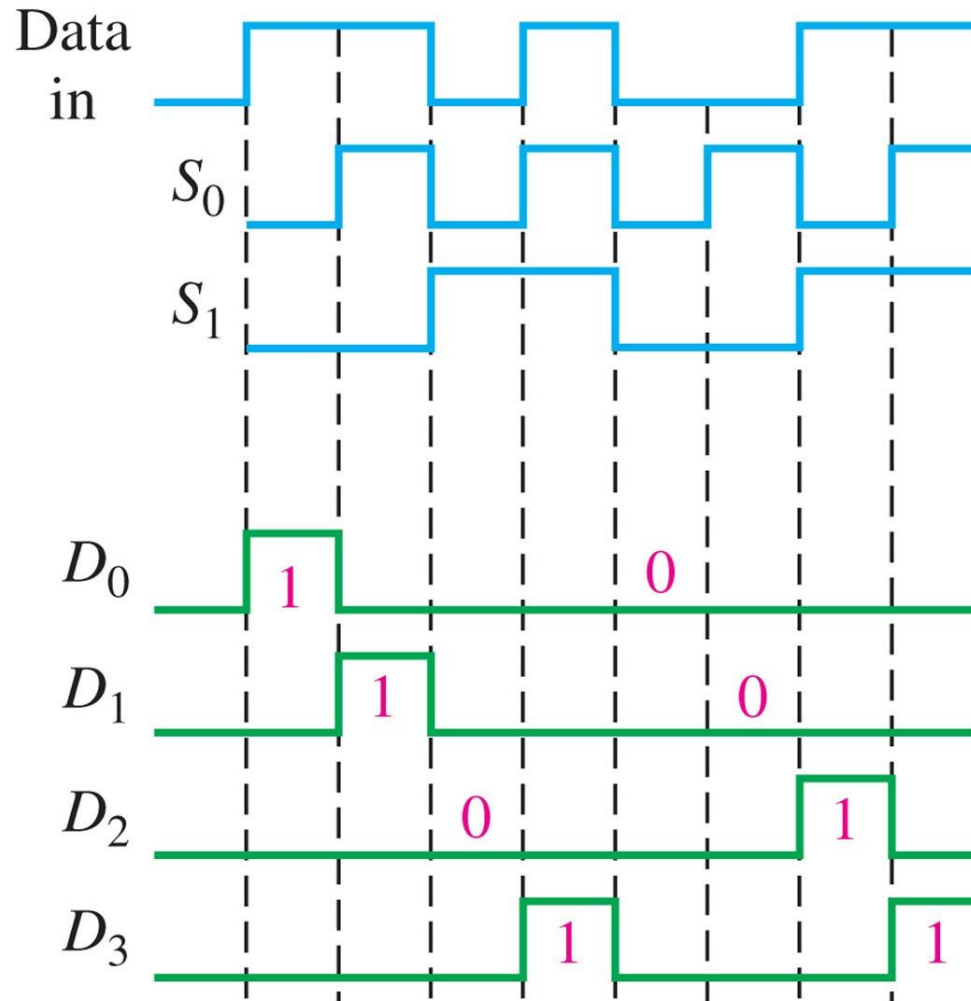
Demultiplexers (Data Distributors)

Exercise: Design the circuit of 4 bit demultiplexer



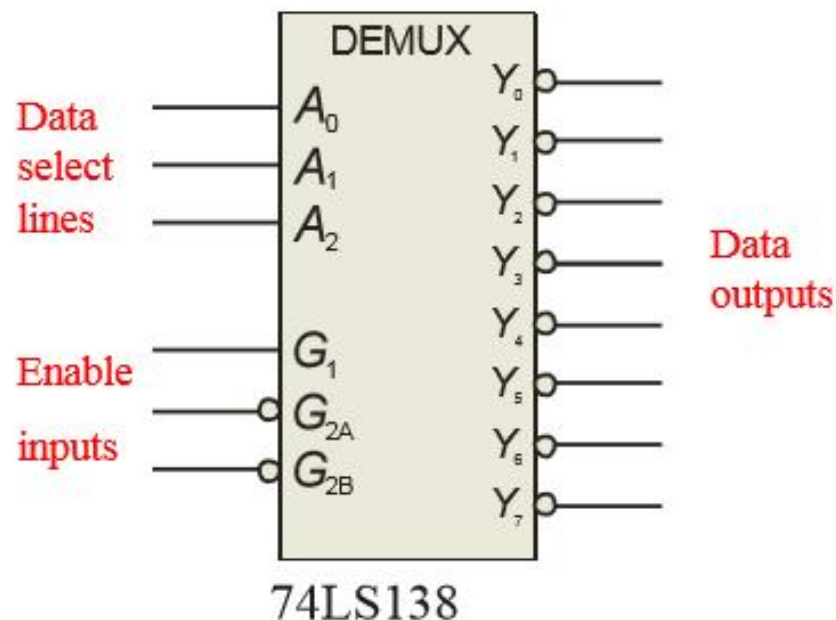
Demultiplexers (Data Distributors)

S0, S1:
High Active



Demultiplexers (Data Distributors)

The 74LS138 was introduced previously as a decoder but can also serve as a DEMUX. When connected as a DEMUX, data is applied to one of the enable inputs, and routed to the selected output line depending on the select variables. Note that the outputs are active-LOW as illustrated in the following example...



Demultiplexers (Data Distributors)

7.3 Device Functional Modes

Table 7-1. Function Table 'HC138, 'HCT138

INPUTS						OUTPUTS							
STROBE			ADDRESS										
G2	$\overline{G1}$	$\overline{G0}$	A ₂	A ₁	A ₀	Y ₀	Y ₁	Y ₂	Y ₃	Y ₄	Y ₅	Y ₆	Y ₇
X	X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	L	H	H	H	H	H	L	H	H	H	H
H	L	L	H	L	L	H	H	H	H	L	H	H	H
H	L	L	H	L	H	H	H	H	H	H	L	H	H
H	L	L	H	H	L	H	H	H	H	H	H	L	H
H	L	L	H	H	H	H	H	H	H	H	H	H	L

Selected $Y = \overline{G0}$ if $\overline{G1} = L$ and $G2 = H$

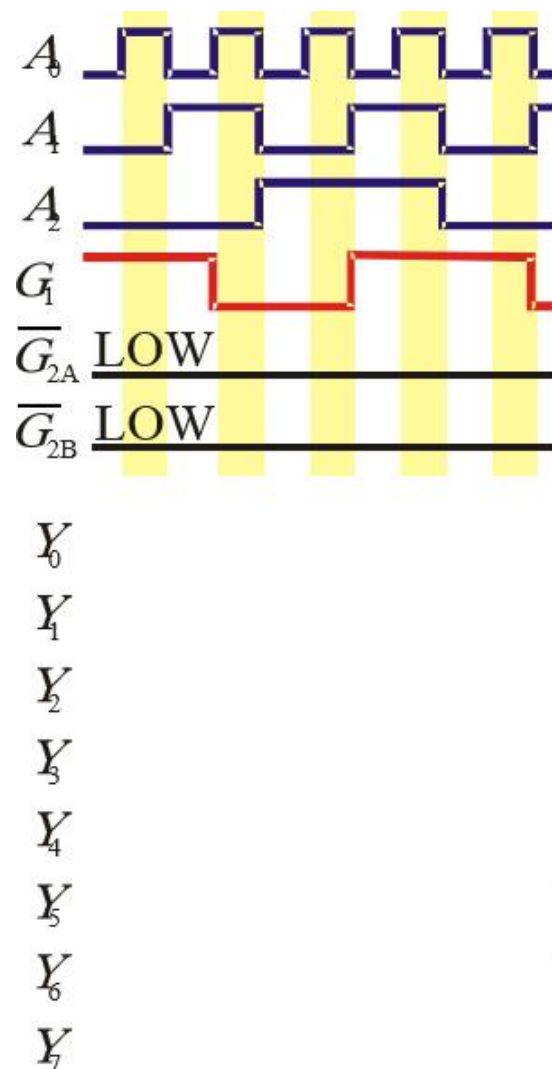
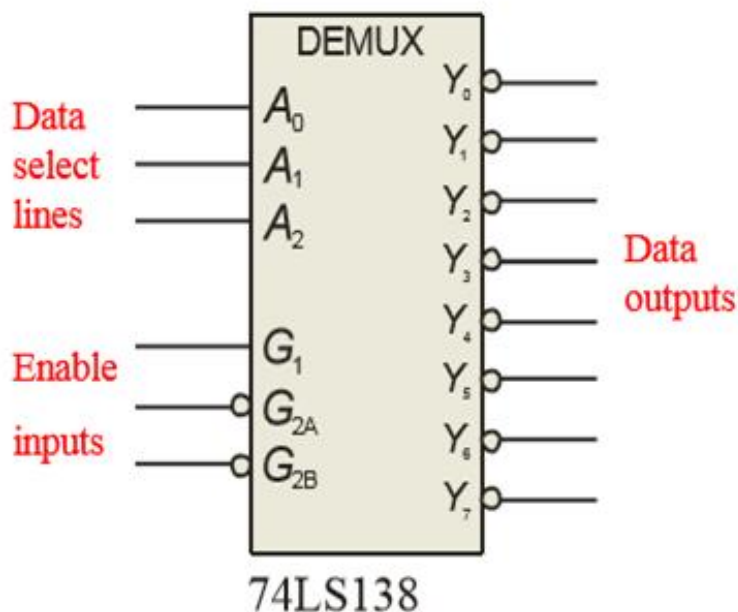


Demultiplexers (Data Distributors)

Example Solution

Determine the outputs, given the inputs shown.

The output logic is opposite to the input because of the active-LOW convention. (Red shows the selected line).

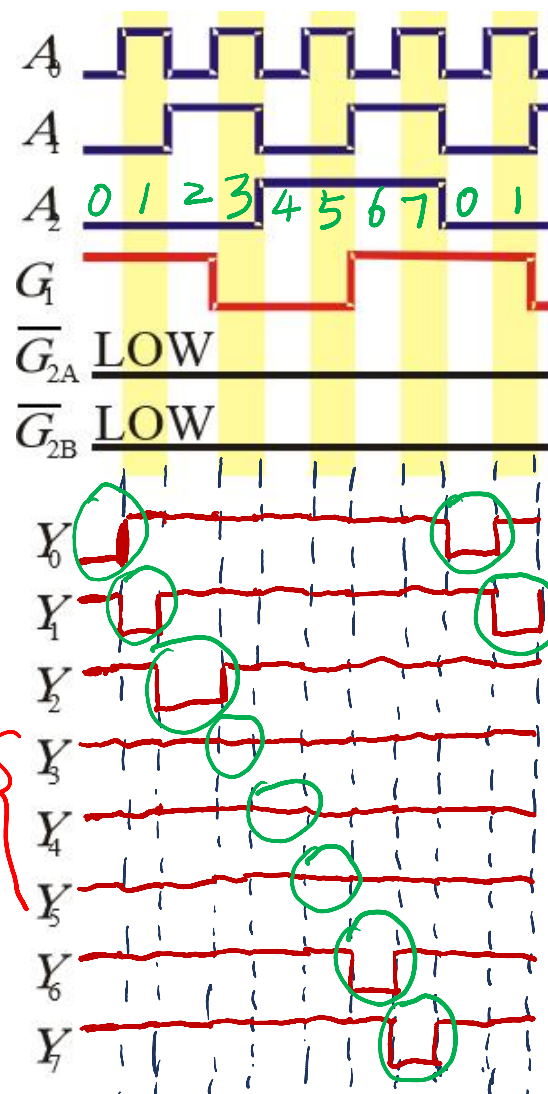
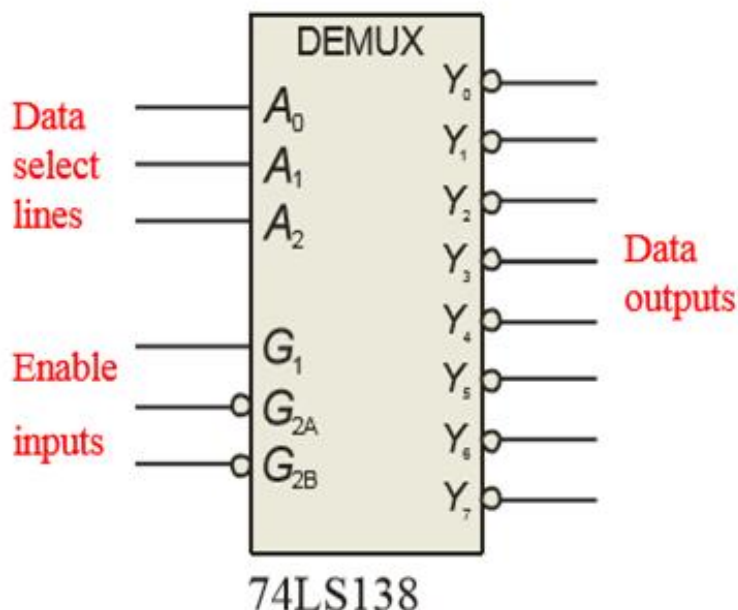


Demultiplexers (Data Distributors)

Example Solution

Determine the outputs, given the inputs shown.

The output logic is opposite to the input because of the active-LOW convention. (Red shows the selected line).



A hazard refers to an unintended transient glitch at the output caused by unequal propagation delays along different signal paths.

1. Static hazard

The output is supposed to remain constant, but it glitches.

2. Dynamic hazard is:

A condition where the output of a logic circuit is supposed to change from one logic value to another, but instead of making a single transition, it toggles multiple times before settling.

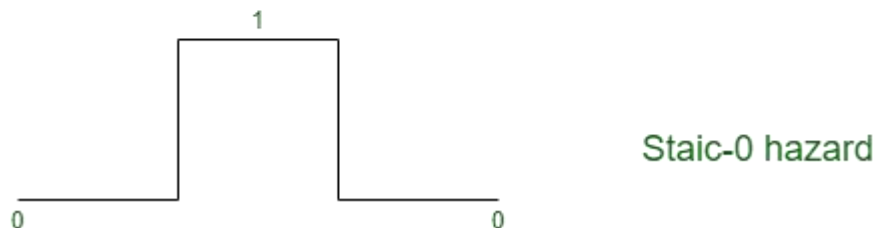
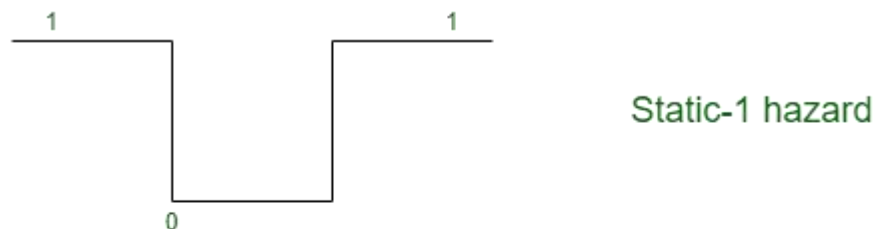
Static Hazard

- Static-1 hazard (state-1 hazard)

Output should remain 1, but briefly goes: $1 \rightarrow 0 \rightarrow 1$

- Static-0 hazard (state-0 hazard)

Output should remain 0, but briefly goes: $0 \rightarrow 1 \rightarrow 0$

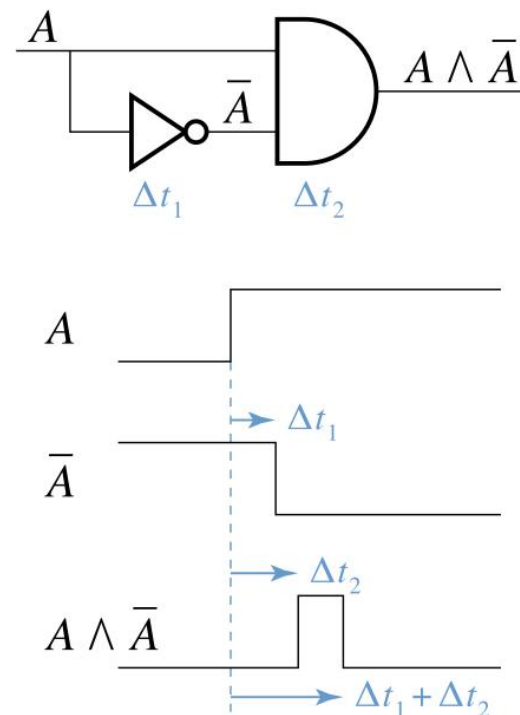


Static Hazard

Example:

Hazard caused by “race condition”

A and A' does not invert at the same time perfectly .



$$F=A+A'B$$

Does a static hazard exist?

If yes, how to fix it?

Hint: inverting A will concurrently change two terms in different paths.

In that case, add a “secure” term not affected by A (while keep F unchanged) may fix this problem.



$$F=A+A'B$$

Does a static hazard exist?
If yes, how to fix it?

Step 1: Normalize to SOP form. Look for complementary variables.
e.g., if it contains A and A': possible hazard, especially when B=1



Static Hazard

Step 2: Test for hazard.

When $B=1$, $F=A+A'$. This means: if A from 0 to 1:

First term $A=1$ ($0 \rightarrow 1$), $A'B=1$ ($1 \rightarrow 0$), thus $F=1$ ($1 \rightarrow 1$)

If A and $A'B$ does not transit concurrently, static hazard occurs (type 1).

Step 3: Fix it

Consider theorem: $AB+A'C = AB+A'C+BC$.

(BC is not affected by A !)

hazard-free expression: $F= A+A'B=A+A'B+B$

or simply $F=A+B$

